Machine Learning Project

Student Roll Number: [00800243]

Module:

Machine Learning

Classification Task:

Predicting Annual Income Category Using Machine Learning

Regression Task:

Predicting Weekly Working Hours Using Machine Learning

University of Salford

Word Count : 2358

1. Problem Framing and Machine Learning Workflow

Machine learning is increasingly being used to support data driven decision making across a wide range of sectors (employment, finance, public policy, workforce planning etc). This project makes use of supervised machine learning on the Adult Census Income dataset to perform two separate prediction tasks. The first is a classification task in which we are attempting to predict whether an individual makes over $50k per year. The second is a regression task in which we are attempting to predict the number of hours worked per week given a range of demographic, education and employment attributes.

These prediction tasks were picked because income and working patterns are all impacted by a whole host of socio-economic and demographic factors. They're a fantastic way of getting experience in the full ML lifecycle and a clear demonstration of the ability to take raw data and turn it into something useful.

The project has proceeded according to the CRISP-DM (Cross Industry Standard Process for Data Mining) methodology which consists of the following stages: Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation and Deployment. The first stage was some exploratory analysis to get a feel for the data and to identify any data quality issues. The data was then cleaned and prepared via the processes of cleaning, feature preparation, encoding and scaling. A number of models were then built for both the classification and regression problems. Model performance was then assessed using the appropriate metrics, the results were interpreted via feature importance and the models were also examined from an ethical and fairness perspective.

The classification task was assessed in terms of Accuracy, Precision, Recall and F1 Score. The regression task was assessed in terms of MAE, RMSE and R^2.

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2. Data and Exploratory Data Analysis

The Adult Census Income dataset was selected as it has both numeric and categorical variables and is appropriate for both classification and regression. It has 32,561 records and 15 attributes which describe demographic characteristics, level of education, employment and income.

Initial analysis identified 6 numeric and 9 categorical attributes. On inspection of the data we spotted some hidden missing values. These were predominantly in the workclass, occupation and native. Country fields and were treated as missing during the pre-processing.

EDA was carried out in order to look at the data for any patterns, distributions and relationships prior to model development. The income distribution analysis revealed that the data is quite imbalanced with a large number of individuals in the dataset earning $50k or less. This meant that in addition to Accuracy we would need to consider Precision, Recall and F1 Score when assessing the performance of the model.

The age distribution has the bulk of the observations in the 'normal' working age groups so age is very likely to be a significant predictor in both of the modelling exercises. There is a clear difference in working hours between the lower and higher income groups which would suggest that the labour force participation variables will be important in terms of their predictive ability.

Overall the exploratory analysis was very useful in terms of informing our decisions around preprocessing and model selection. The dataset was a mix of numerical and categorical variables with missing values and a class imbalance which is pretty much the classic real world dataset scenario.

Figure 1: Income Distribution

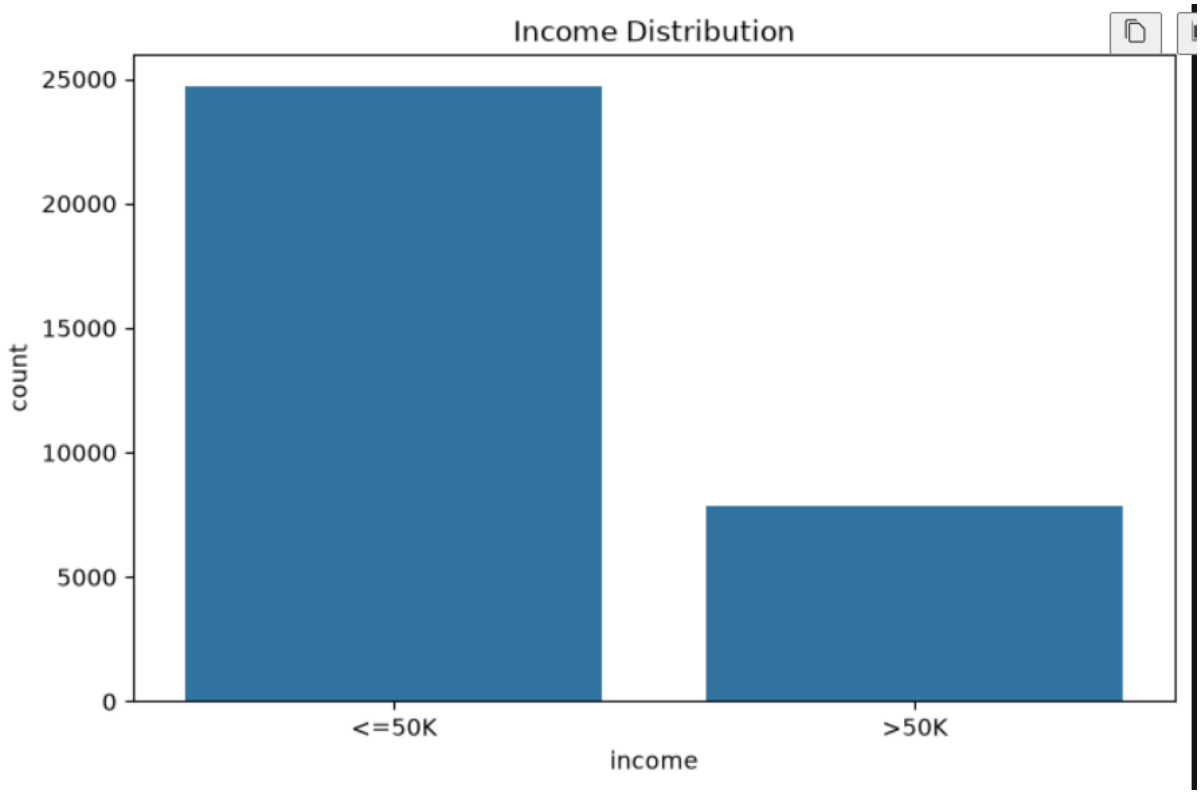


Figure 2: Age Distribution

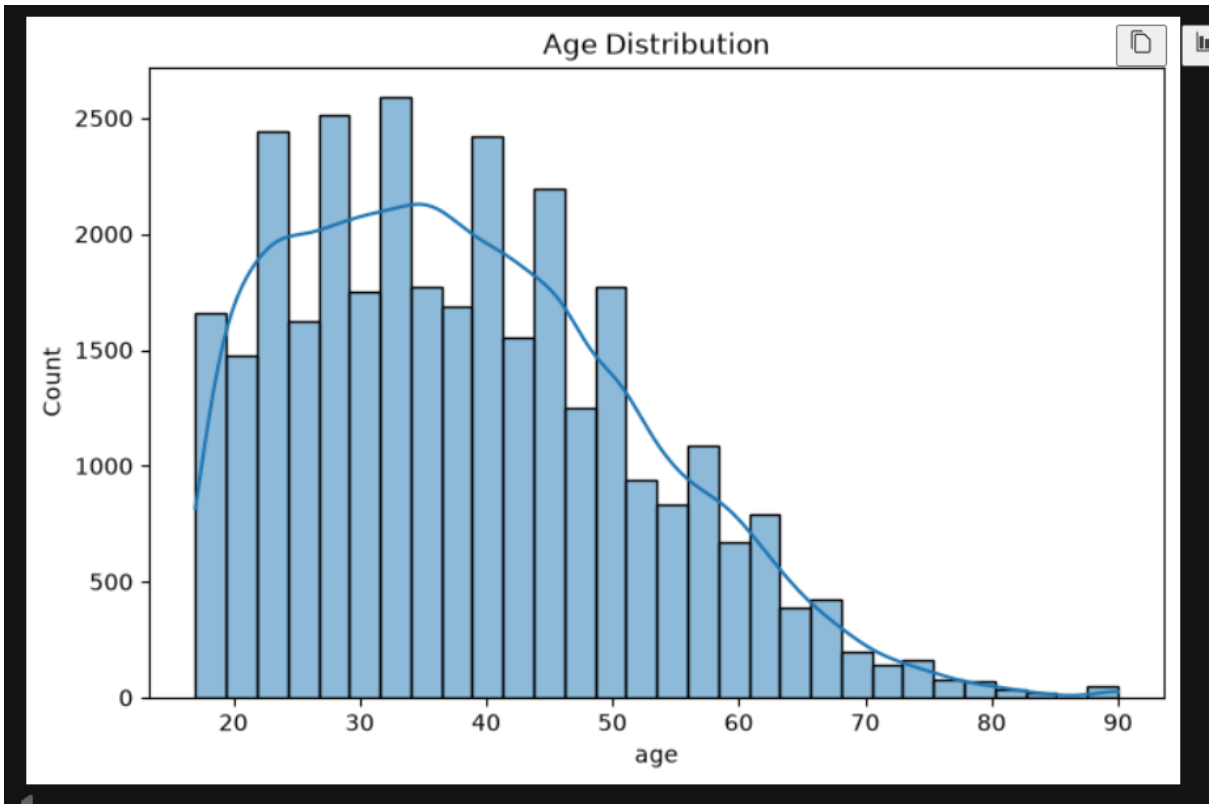
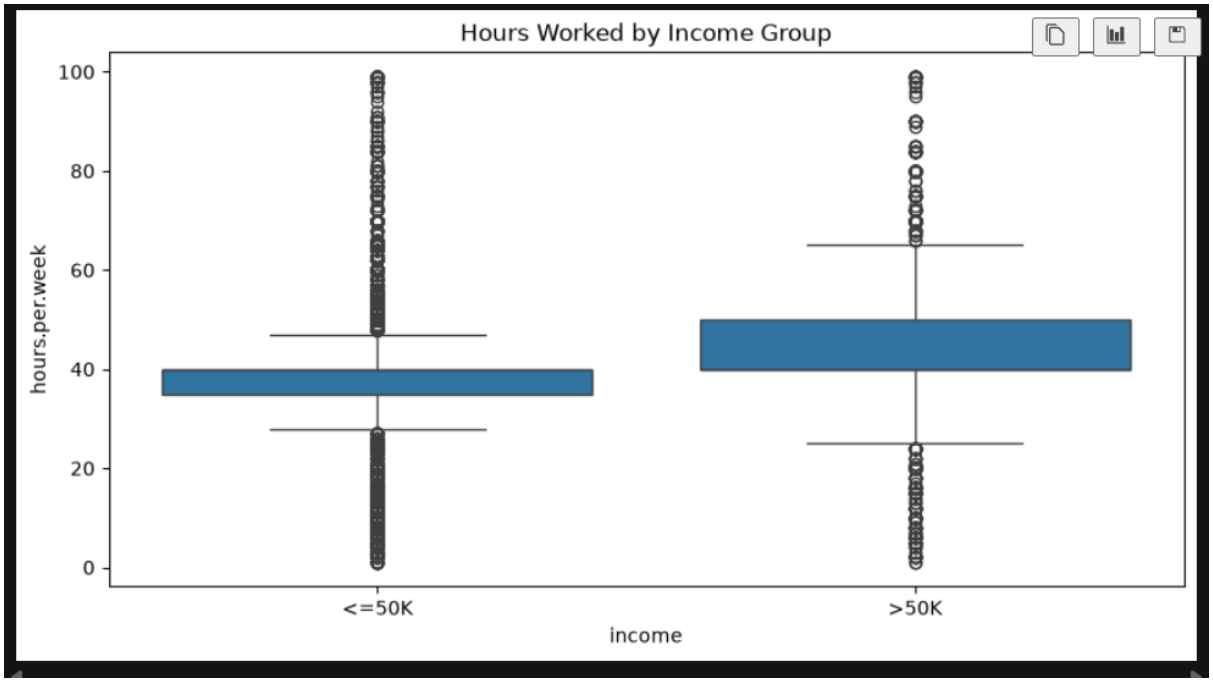


Figure 3: Hours Worked by Income Group



3. Data Preprocessing and Feature Engineering

Data pre-processing was performed to get the data into a form suitable for the machine learning algorithms. The hidden missing values (denoted by the ? character) were first replaced with nulls and then dealt with via imputation.

The dataset was split into two modelling tracks. One was a classification track (target variable was annual income) and the other was a regression track (target variable was number of hours worked per week). The training and test sets were created using an 80/20 split. For the classification track we used stratified sampling to preserve the distribution of the target.

Features were split into numerical and categorical. The numerical features were age, final weight, education number, capital gains and capital losses. The categoricals were workclass, education, marital status, occupation, relationship status, race, sex and native country.

To ensure reproducibility and to avoid any data leakage we used Scikit-Learn pipelines for the preprocessing. The numeric variables were imputed using the median and then standardised using StandardScaler. The categorical variables were imputed using the mode and then one-hot encoded.

A ColumnTransformer was used to combine all of the above preprocessing into a single pipeline so that all of the ML algorithms were getting the same processed input data. That was clearly a step in the right direction in terms of maintainability, reproducibility and following proper ML development practice.

By automating the pre-processing via pipelines we've hugely reduced the risk of any inconsistency between the training and the test sets and at the same time improved the efficiency of the workflow.

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4. Classification Modelling

Four ML algorithms were tested for the classification task: Logistic Regression, Decision Tree, Random Forest and XGBoost.

Logistic Regression was the baseline classification model (chosen for its simplicity, interpretability and ubiquity in the binary classification context). It achieved an Accuracy of 85.3% and an F1 Score of 66.4% which is a very solid baseline against which to compare the other models.

The Decision Tree classifier was used to see if the non-linearity in the data would lead to any improvement in performance. It achieved an Accuracy of 85.2% and an F1 Score of 66.1%. Decision Trees are of course very interpretable in that they give you a very clear set of decision rules, but in this case they didn't perform significantly better than the Logistic Regression.

Random Forest was the chosen ensemble method as they generally lead to improved predictive accuracy through the reduction of overfitting. It delivered an accuracy of 85.1% and an F1 score of 66.5%. Recall was marginally better than with the Logistic Regression model but the overall performance gain was still very small.

XGBoost was finally implemented as a gradient boosting algorithm which is able to capture complex relationships in structured tabular data. It delivered the best performance on all the evaluation metrics. It achieved an Accuracy of 86.9%, a Precision of 77.6%, a Recall of 64.0% and an F1 Score of 70.2%.

The results show that gradient boosting was better at picking up the interactions between the demographic and employment variables than either of the simpler linear or tree models. XGBoost is therefore the model of choice for classification.

Confusion Matrix (XGBoost)

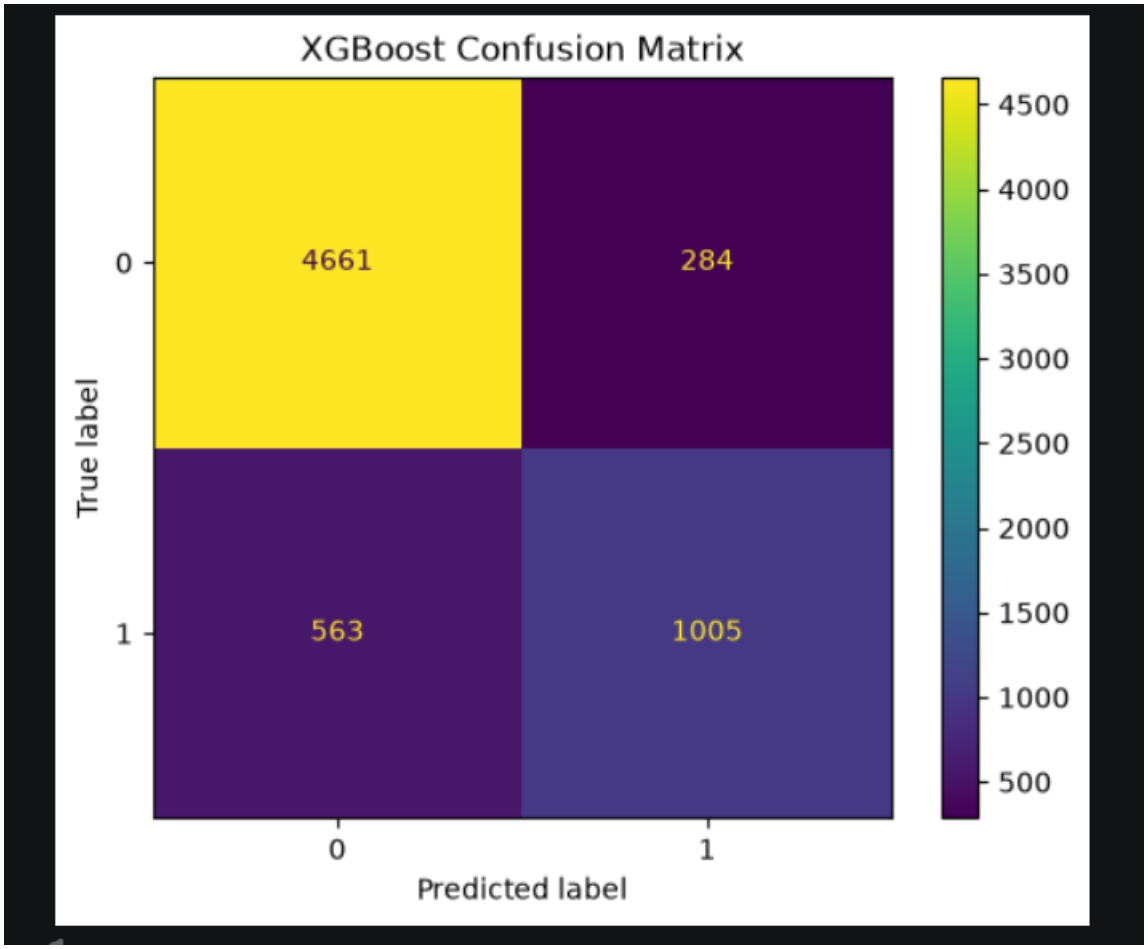


Table 1: Classification Model Comparison

	Model	Accuracy	Precision	Recall	F1
3	XGBoost	0.8689	0.7759	0.6403	0.7016
2	Random Forest	0.8512	0.7260	0.6135	0.6651
0	Logistic Regression (L2)	0.8531	0.7392	0.6020	0.6636
1	Decision Tree	0.8521	0.7373	0.5995	0.6613

5. Regression Modelling

Five regression algorithms were developed and tested: Linear Regression, Ridge Regression, Decision Tree Regressor, Random Forest Regressor and XGBoost Regressor.

Linear regression was the baseline model and achieved an MAE of 7.78 hours and an RMSE of 11.12 hours. The R^2 was 0.196 so approximately 19.6% of the variation in weekly working hours was explained by the model.

Ridge Regression was used to see if regularisation would in fact lead to any improvement in generalisation. The performance was pretty much the same as Linear Regression (very slight improvement in R^2 at 0.196) and the errors were very similar.

The Decision Tree Regressor was used to model the potential for non-linearity. It had a very slightly better MAE (7.67 hours) but a dramatically worse R^2 (0.172) so in the overall it was a much weaker model.

Random Forest regression was then tried to see if ensemble learning would do any better. It gave an MAE of 7.75 hours and an R^2 of 0.188. It did at least in some small way mitigate the problems with individual decision trees, but the improvement overall was very slight. XGBoost Regressor delivered the best regression performance. Lowest MAE (7.31 hours), lowest RMSE (10.62 hours) and highest R^2 (0.266). The explanatory power is still very modest, but the results do suggest that the gradient boosting approach has picked up more of the complex relationships than any of the other approaches. Hence XGBoost Regressor was the selected regression model.

Table 2: Regression Model Comparison

	Model	MAE	RMSE	R2
0	Linear Regression	7.78	11.12	0.196
1	Ridge Regression	7.78	11.12	0.196
2	Decision Tree Regressor	7.67	11.28	0.172
3	Random Forest Regressor	7.75	11.18	0.188
4	XGBoost Regressor	7.31	10.62	0.266

XGBoost Regressor delivered the best overall regression performance. It had the lowest MAE (7.31), the lowest RMSE (10.62) and the highest R^2 (0.266). That said it was clearly the most effective at capturing the underlying complexity of the data as compared to Linear Regression, Ridge Regression, Decision Tree Regression and Random Forest Regression.

6. Evaluation and Validation

Model evaluation was done using both a held out test set and cross-validation. XGBoost was by far the best for the classification task (86.9% accuracy, 70.2% F1). Logistic Regression was also very good. The Decision Tree and the Random Forests were a bit worse overall. To assess model robustness we performed 5 fold stratified cross-validation on the XGBoost classifier. The model achieved a mean cross-validation accuracy of 87.27% (std dev 0.0027). The extremely low standard deviation across the folds is a very good indicator of stability and reliable generalisation performance. For the regression task the XGBoost Regressor was by far the best performing with an MAE of 7.31, an RMSE of 10.62 and an R^2 of 0.266. Five fold CV gave a mean R^2 of 0.273 and a standard deviation of 0.0128.8.The validation results point to both XGBoost models generalising very well and being fairly insensitive to a particular train/test split. That further reinforces our confidence in the reported performance.

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7. Model Interpretation

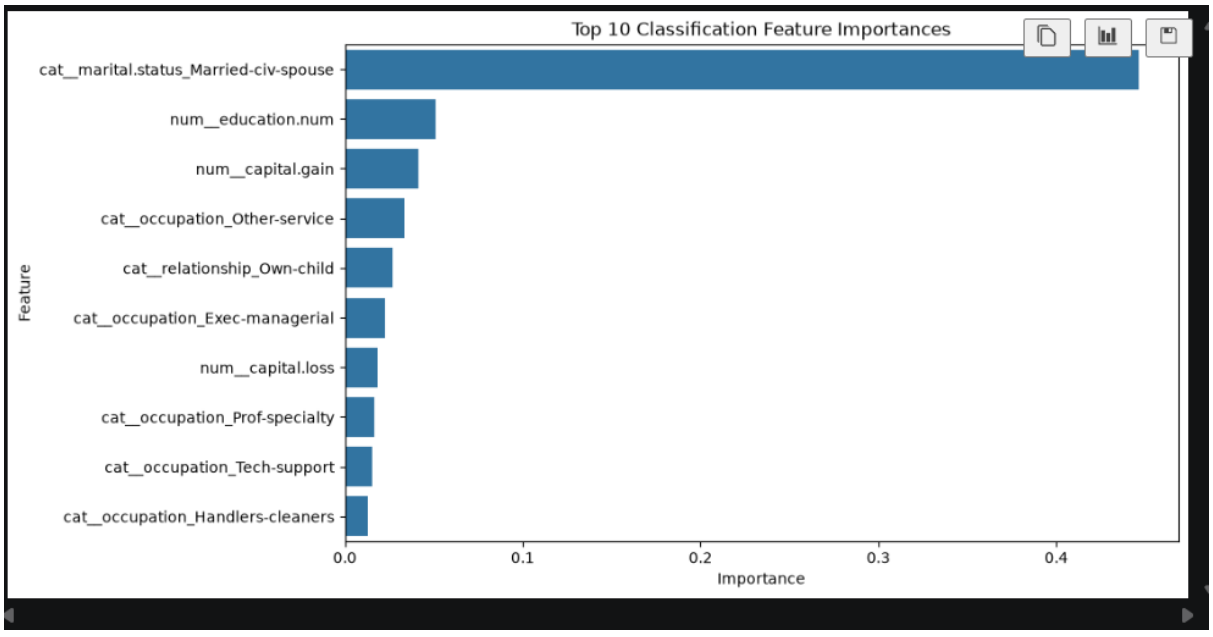
Interpretability is a key part of machine learning as it enables us to understand how the predictions are being made and to verify that the relationships that the model has learned are sensible in the real world. The XGBoost classification model's feature

importance analysis showed that marital status was by far the most important feature in determining the income group. Education, capital gains, occupation and relationship were all fairly important features. Nothing surprising there really given the nature of the data and the way we'd expect income to be related to education, employment etc.

For the regression task the feature importance analysis revealed that sex, occupation, income bracket, age and relationship status all made a significant contribution to the prediction of weekly working hours. Once again it's quite different from the classification model in that the importance values were spread across a number of variables rather than one variable being by far the most important.

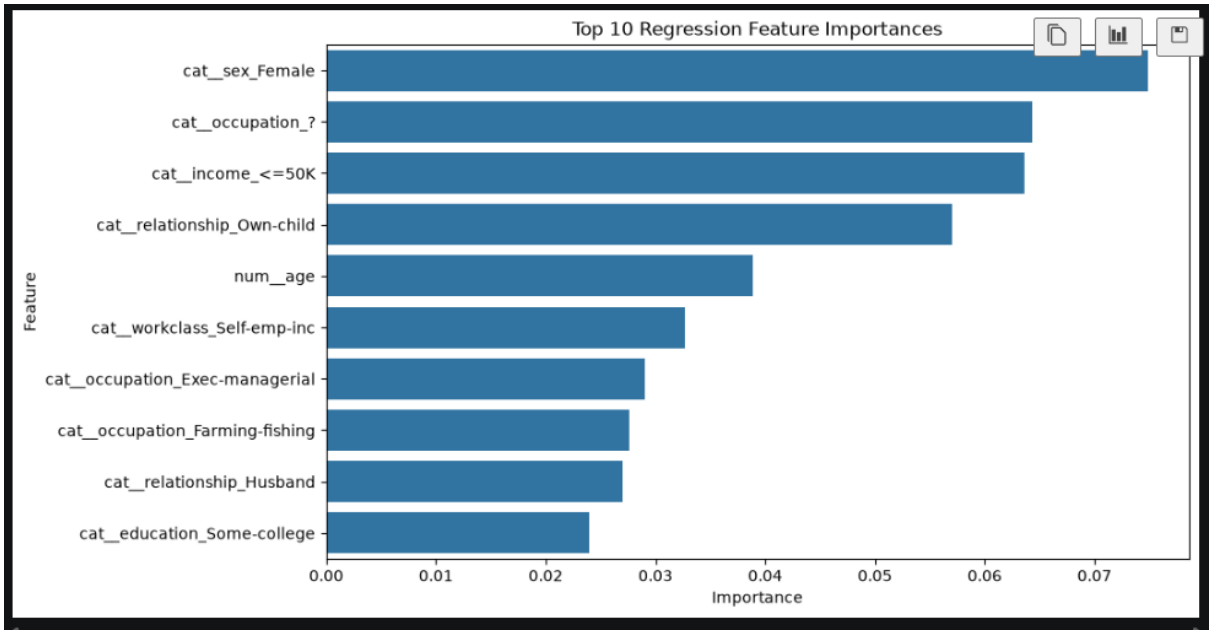
The feature importance results are a huge improvement in terms of transparency as they identify the features that had the greatest influence on the model's behaviour. That's a massive help in terms of explainability and means that the model's behaviour is something that can be understood by both technical and non-technical users.

Figure 5. Top 10 Classification Feature Importances



The feature importance analysis pointed to marital status, education, capital gains, occupation and relationship status as the most important factors in predicting income. That makes perfect sense in a socioeconomic context. The model has clearly learned the relevant patterns.

Figure 6. Top 10 Regression Feature Importances



The regression feature importance analysis revealed that sex, occupation, income category, age and relationship status were all amongst the most important predictors of weekly working hours. It's quite different to the classification model in that importance is spread across several variables rather than being all in one. This indicates that weekly working hours are the result of a number of different factors. The analysis does a lot to improve the transparency of the model in that it identifies the variables which are most important in determining the outcome.

8. Fairness and Ethics

The Adult Census Income dataset contains sensitive demographic attributes (sex and race) and the use of those variables is fraught with ethical issues as they will necessarily reflect the existing societal inequalities.

Feature importance analysis did in fact identify sex as a significant predictor in the regression model. That might be a good thing in terms of improving the predictive accuracy of the model, but it does raise the issue of potential bias and fairness. All the usual suspects in terms of bias (historical, sampling and representation) could be in play and could potentially lead to some groups being disadvantaged if the model were to be used in a real world decision making scenario.

A frequently used fairness metric is the Demographic Parity Difference which looks at whether the positive outcomes are equally distributed across the different demographic groups. Future work will be to use Fairlearn to look at the prediction outcomes across the sex and race groups and assess whether any disparities are present.

Compliance with the GDPR principles is also of the essence when it comes to the handling of personal data. The systems need to be transparent, the processing needs to be lawful, there needs to be accountability and data minimisation. High predictive performance in itself is not enough to ensure responsible deployment.

Fairness, explainability and data protection are all factors that must be considered as part of the technical evaluation when you're building ML systems.

9. Reflection and Future Work

This project has given us a tremendous amount of hands-on experience of the full machine learning workflow from the exploratory analysis and preprocessing through to model evaluation and interpretation.

One of the key takeaways was the importance of data prep. How you deal with missing values, how you encode your categoricals and how you put together your pre-processing pipelines has a massive impact on both model performance and reproducibility. The project also demonstrated that more complex models don't necessarily do better. The Decision Tree and Random Forests gave fairly good results but XGBoost was by far the best in both classification and regression.

Future work could involve a much more rigorous hyperparameter optimisation via GridSearchCV, the implementation of SHAP explanations to improve interpretability and the application of Fairlearn to perform a full fairness analysis. The models that emerge as the best performers could then be deployed via a web app framework (e.g. Streamlit or Flask) to enable real time prediction and user interaction.

Overall the project has been a success in demonstrating the application of ML to both classification and regression problems and has underlined the importance of evaluation, interpretability and responsible AI.

References (APA 7)

Bache, K., & Lichman, M. (2013). *Adult Data Set*. UCI Machine Learning Repository. https://archive.ics.uci.edu/ml/datasets/adult

Breiman, L. (2001). Random forests. *Machine Learning, 45*(1), 5–32. https://doi.org/10.1023/A:1010933404324

Chapman, P., Clinton, J., Kerber, R., Khabaza, T., Reinartz, T., Shearer, C., & Wirth, R. (2000). *CRISP-DM 1.0: Step-by-step data mining guide*. SPSS Inc.

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). Association for Computing Machinery. https://doi.org/10.1145/2939672.2939785

European Parliament and Council of the European Union. (2016). *Regulation (EU) 2016/679 (General Data Protection Regulation)*. Official Journal of the European Union. https://eur-lex.europa.eu/eli/reg/2016/679/oj

Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Annals of Statistics, 29*(5), 1189–1232. https://doi.org/10.1214/aos/1013203451

Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Passos, A., Cournapeau, D., Brucher, M., Perrot, M., & Duchesnay, E. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research, 12*, 2825–2830.

Sarker, I. H. (2021). Machine learning: Algorithms, real-world applications and research directions. *SN Computer Science, 2*(3), 160. https://doi.org/10.1007/s42979-021-00592-x

Shapley, L. S. (1953). A value for n-person games. In H. W. Kuhn & A. W. Tucker (Eds.), *Contributions to the Theory of Games* (Vol. 2, pp. 307–317). Princeton University Press.

United Nations. (2021). *Using artificial intelligence for social good*. United Nations Publications.